A SHEAF WITH A REPRESENTABLE SURJECTIVE ETALE COVER IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- The presentation theorem for algebraic spaces. Theorem theorem-presentation, used at proof lines 1897-1899. Produces the algebraic space F' = U/R.
- Surjectivity of a flat, locally finitely presented representable map. Lemma lemma-surjective-flat-locally-finite-presentation (897-904), used at proof lines 1901-1902. Gives surjectivity of U -> F as a map of sheaves.

Source reference. Original lemma 05VM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7556–7556. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original theorem 02WW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3819–3819. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let F be a sheaf on $(Sch/S)_{fppf}$ such that there exists $U \in \mathrm{Ob}((Sch/S)_{fppf})$ and a map $U \to F$ which is representable, surjective, and étale. Then F is an algebraic space.*

Proof. Set $R = U \times_F U$. This is a scheme as $U \to F$ is assumed representable. The projections $s, t : R \to U$ are étale as $U \to F$ is assumed étale. The map $j = (t, s) : R \to U \times_S U$ is a monomorphism and an equivalence relation as $R = U \times_F U$. By Theorem 02WW the quotient sheaf $F' = U/R$ is an algebraic space and $U \to F'$ is surjective and étale. Again since $R = U \times_F U$ we obtain a canonical factorization $U \to F' \to F$ and $F' \to F$ is an injective map of sheaves. On the other hand, $U \to F$ is surjective as a map of sheaves by Lemma 05VM. Thus $F' \to F$ is also surjective and we conclude $F' = F$ is an algebraic space. □

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.